

ENHANCED PROFESSIONAL STUDENT LOGBOOK

Student Name: Haron Mutai

Admission Number: BIT/2024/73474

School/Faculty: Computing& Informatics

Programme: Information Technology

Academic Year: 2026

Semester: Two

Lecturer: Michael Nyoro

Phone Number: 0758345174

Instructions

This logbook is used to document weekly learning activities, practical exercises, skills acquired, challenges encountered, solutions applied, and reflections. Students should complete all sections every week.

Each week to follow remarks in week 1 section.

Autogenerated Table of content

Week 1: Introduction to Mobile Application Development

Practical Activities Undertaken

1. Installed Android Studio.
2. Installed Flutter SDK and Dart SDK.
3. Configured Android Studio with Flutter and Dart plugins.
4. Set up Android Emulator.
5. Created the StarShare project.
6. Created a GitHub repository for project version control.
7. Verified Flutter installation and environment configuration.
8. figure1: Installation and setting up of android studio.



Figure2: Configured Android SDK and Emulator and Verified development environment.

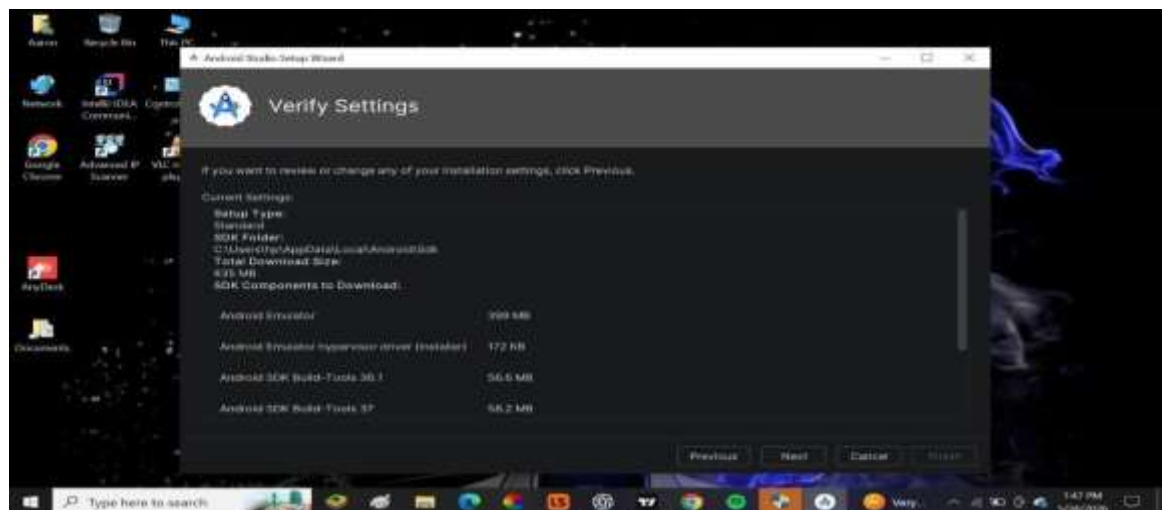
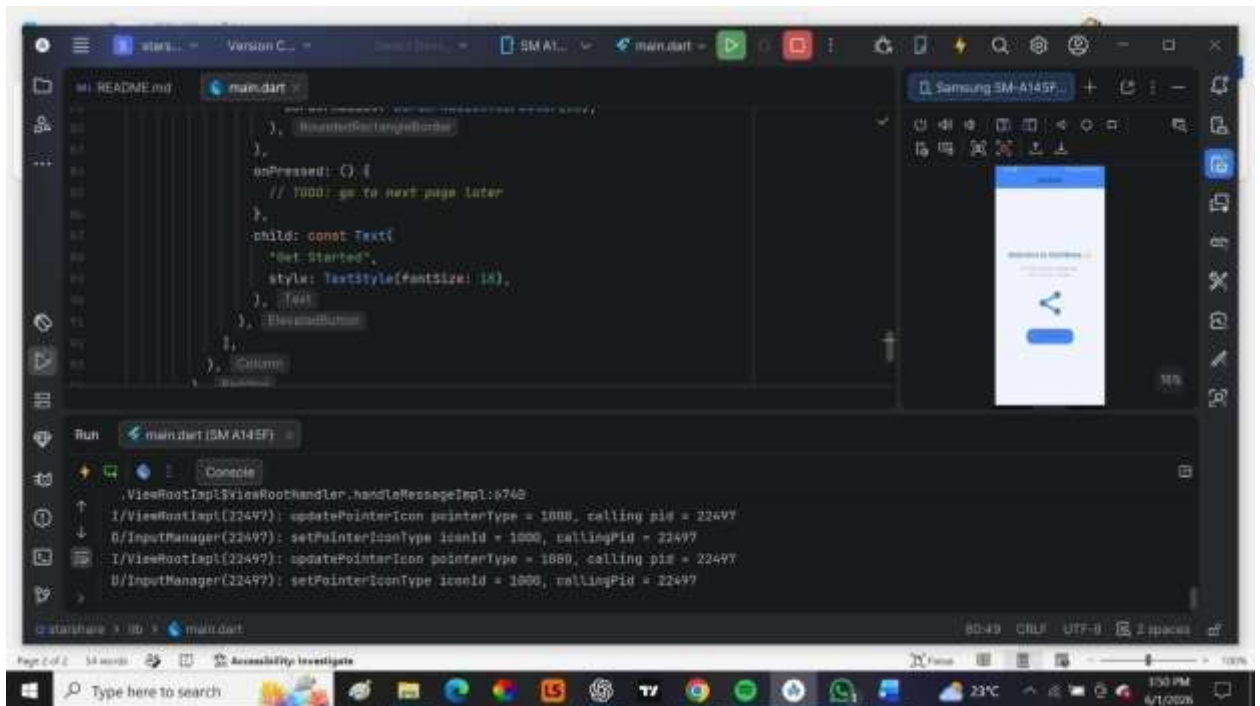
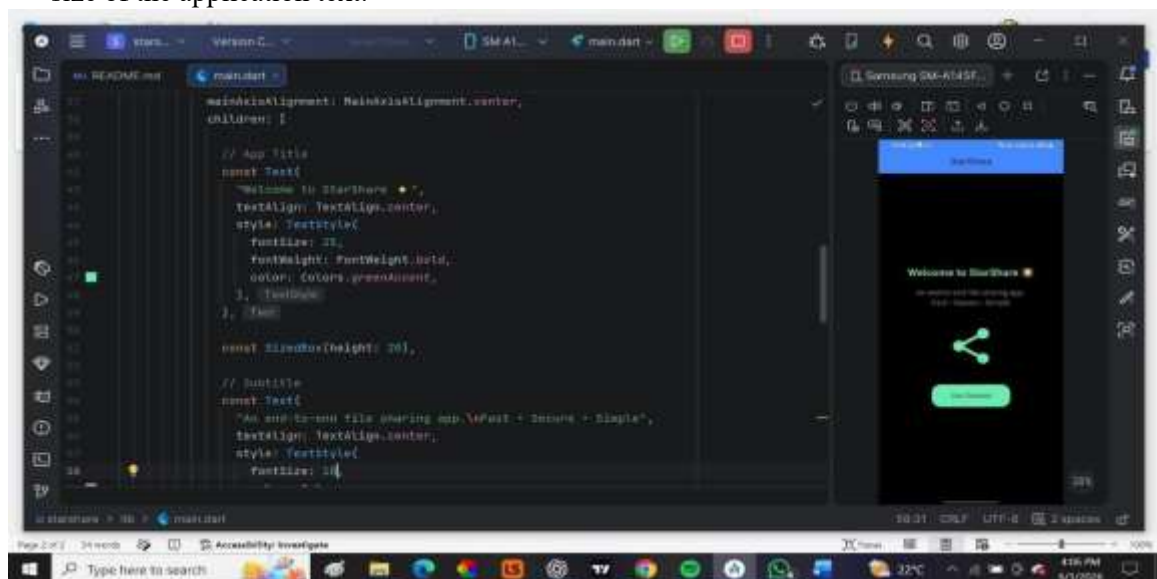


Figure3: Created a new flutter project.



- Figure4: Modified the application title, changed the background colour and adjusted the font size of the application text.



Tools/Software Used

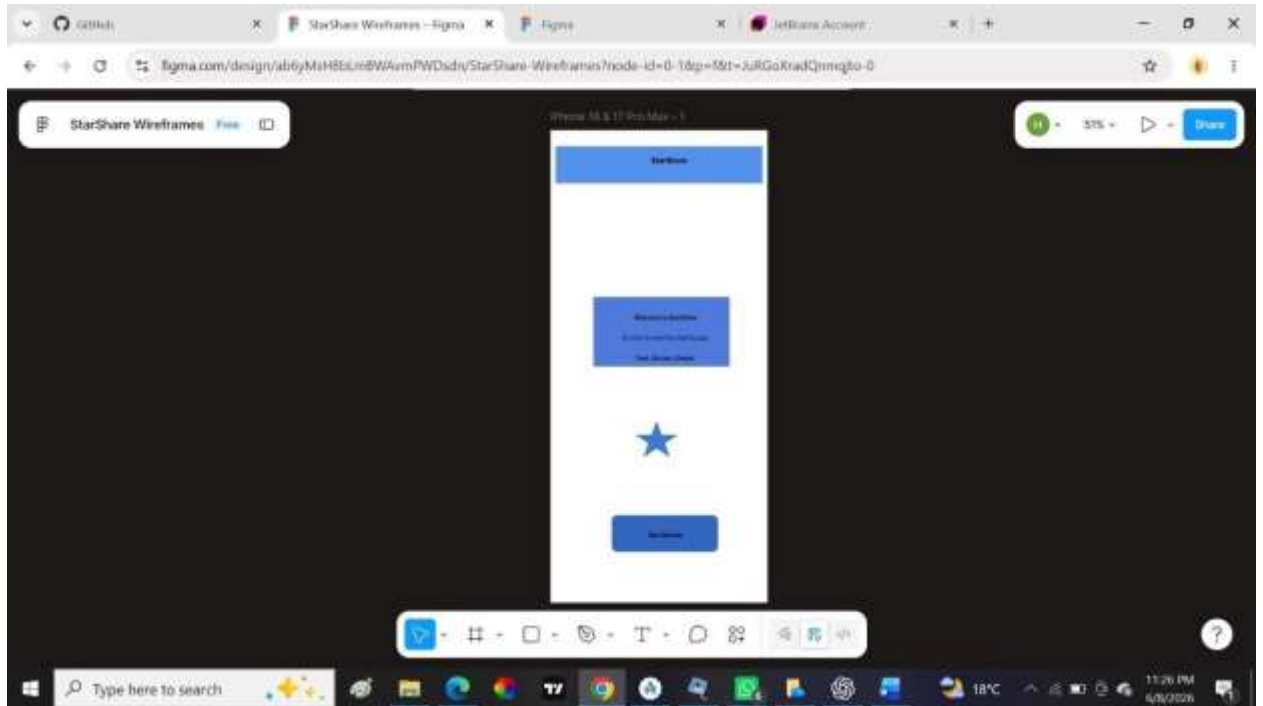
1. Android Studio
2. Flutter SDK
3. Android Emulator
4. Phone
5. Dart SDK
6. Figma

Personal Reflection

I Successfully installed and configured the development environment for the StarShare project. I gained practical experience setting up Flutter, Android Studio, and GitHub, which established a solid foundation for the development process.

Week 2: Mobile Development Environment Practical Activities Undertaken

1. Installed and configured Figma.
2. Explored mobile application UI/UX design principles.
3. Created project wireframe layouts in Figma.
4. Designed Splash Screen wireframe.
5. Designed Welcome/Onboarding Screen wireframe.
6. Designed Login Screen wireframe.
7. Designed Registration Screen wireframe.
8. Reviewed mobile application navigation structures.
9. **Figure 1:** Onboarding Screen Wireframe.



10. **Figure 2:** Splash Screen Wireframe.

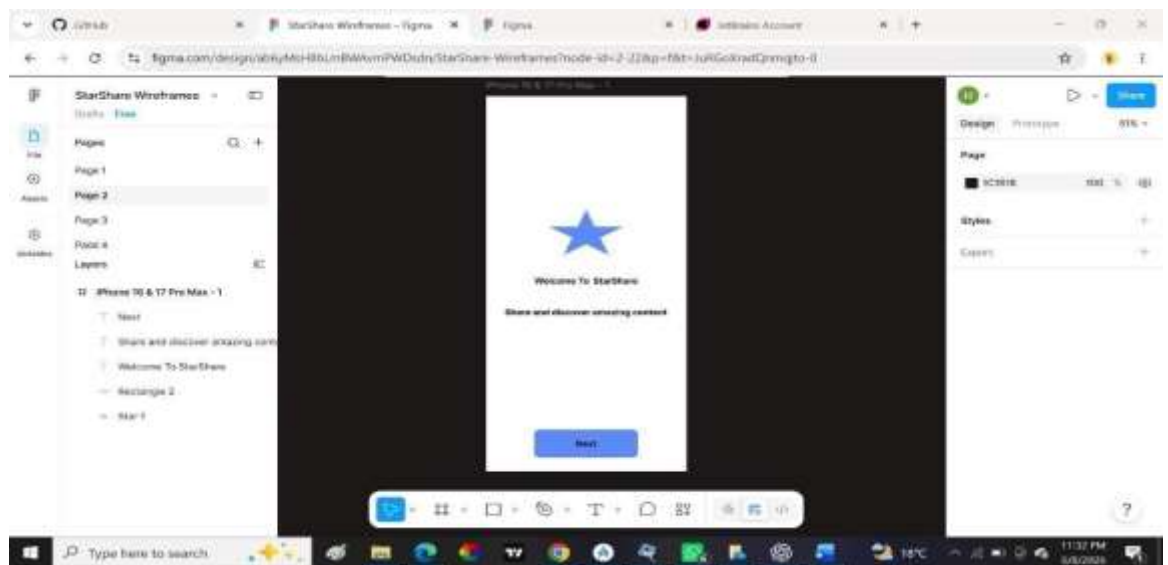


Figure 3: Login screen Wireframe

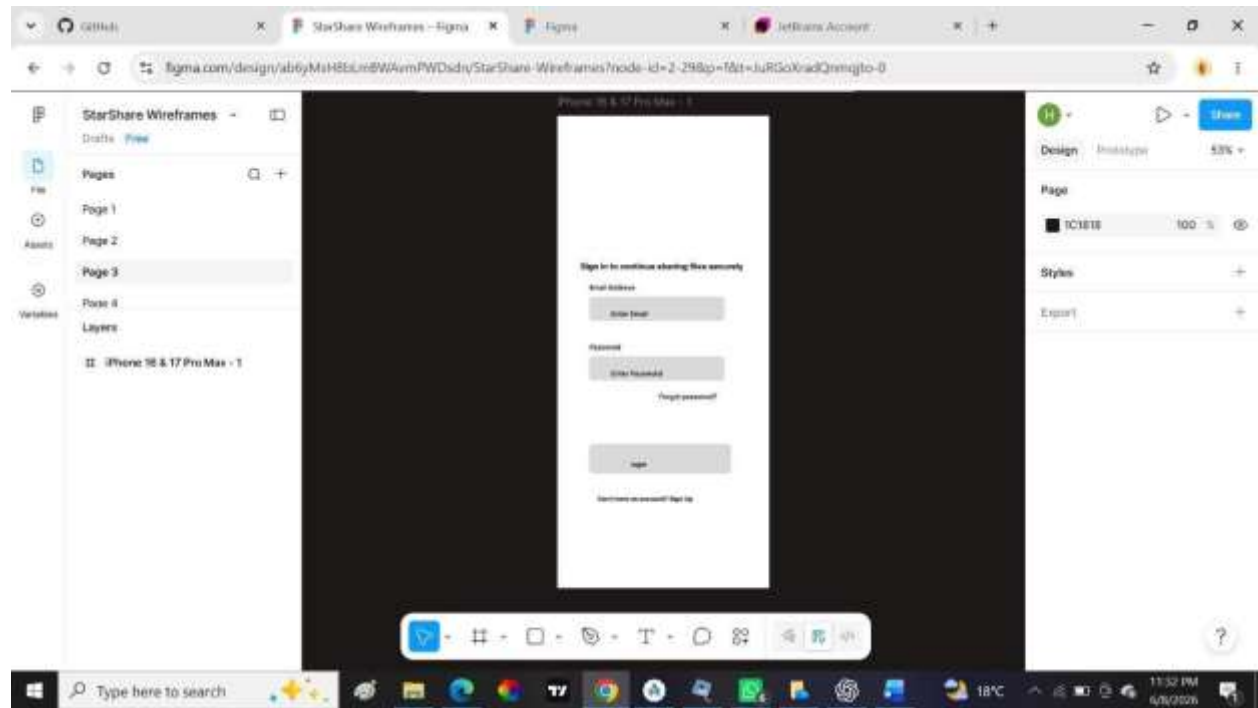
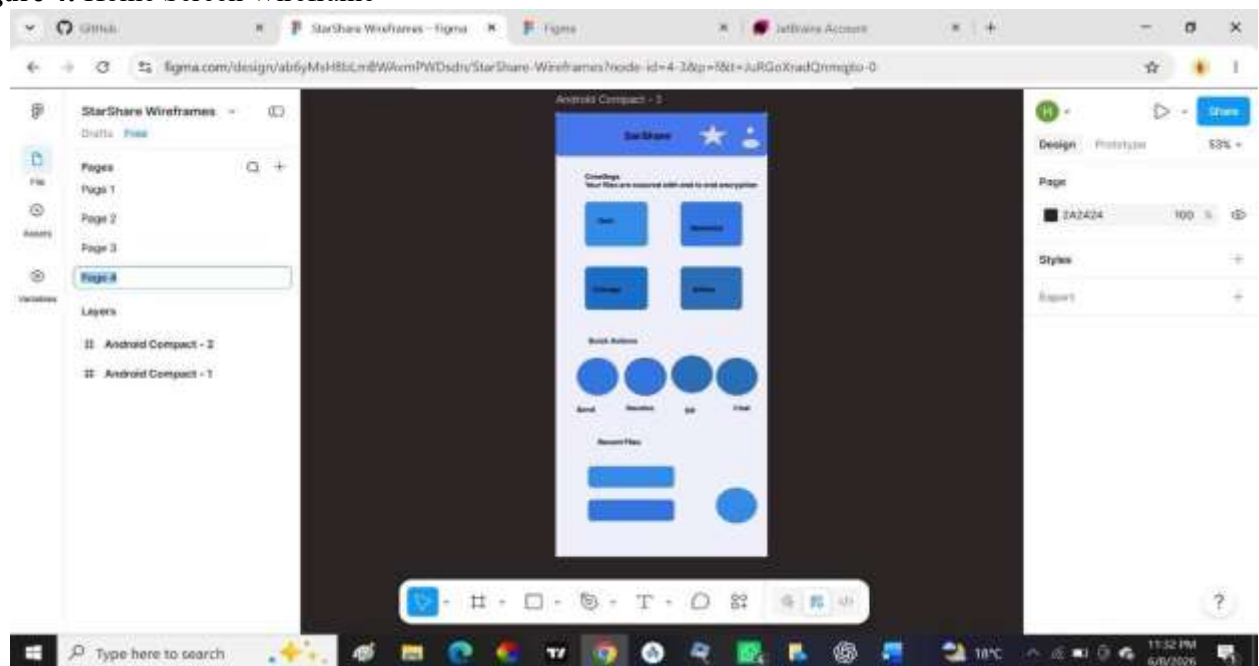


Figure 4: Home Screen Wireframe



Tools used.

- 1.Figma
2. Android studio (for references)

Personal Reflection

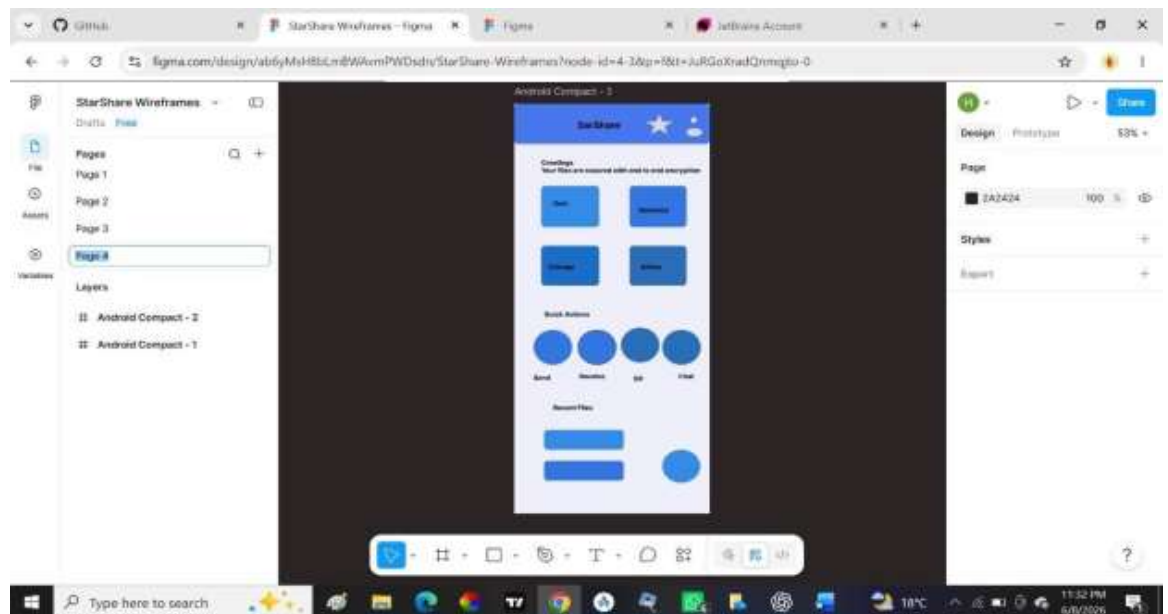
I developed wireframes using Figma and applied UI/UX design principles to improve usability. This activity provided a visual representation of the application structure and guided future development decisions.

Week 3: User Interface Design

Practical Activities Undertaken

1. Refined application wireframes.
2. Designed Home Dashboard interface.
3. Designed File Upload section
4. Designed Storage section
5. Improved navigation flow between screens.
6. Standardized buttons, icons, and layout structures.
7. Reviewed interface consistency and usability.

Figure 1: home screen



Tools/Software Used

1. Figma
2. Android Studio
3. Flutter SDK

Personal Reflection

I refined the user interface design and improved consistency across all screens. The finalized wireframes provided a clear development roadmap and enhanced the overall usability of the StarShare application.

Week 4: Event Handling and User Interaction Practical **Activities Undertaken**

1. Implemented button click events in Flutter UI.
2. Added navigation between screens using routes.
3. Implemented form validation for login and registration.
4. Handled user inputs using TextEditingController.
5. Added loading indicators during user interactions.
6. Implemented basic error handling for invalid inputs.

Figure1



The get started button leads you to onboarding page.

Figure 2



The login button takes you to home page.

Tools used:

1. Flutter SDK
2. Android Studio
3. Dart Programming Language

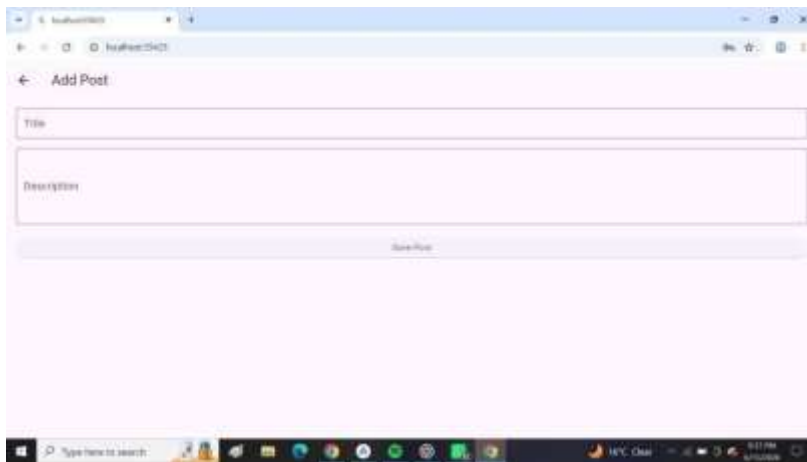
Personal Reflection

I learned how to handle user interactions in Flutter using event listeners, navigation, and form validation. This improved the responsiveness and usability of the application.

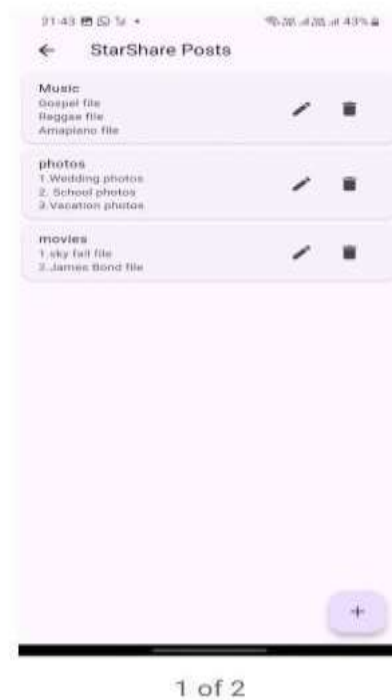
Week 5: Data Management

Practical Activities Undertaken

1. Created data models for users and posts.
2. Implemented local state management using `setState()`.
3. Integrated API service for data fetching and posting.
4. Parsed JSON responses from backend.
5. Displayed dynamic data using `ListView.builder`.
6. Handled loading and error states in data rendering.



Allows adding and saving of posts



This one shows the editing and the deleting part of the system.

Tools Used

1. Flutter SDK
2. Dart
3. API Service (HTTP package)
4. Android Studio

Personal Reflection

I gained practical experience in managing and displaying dynamic data in Flutter applications. I also learned how to integrate backend APIs and handle JSON data effectively.

Week 6: Networking and APIs

Practical Activities Undertaken

1. Created REST API service for backend communication.
2. Implemented HTTP GET and POST requests in Flutter.
3. Connected Flutter application to backend server.

4. Handled API responses and errors.
5. Implemented asynchronous programming using async/await.
6. Tested network connectivity and data flow.

Figure1

This one shows a button for API for backend communication.

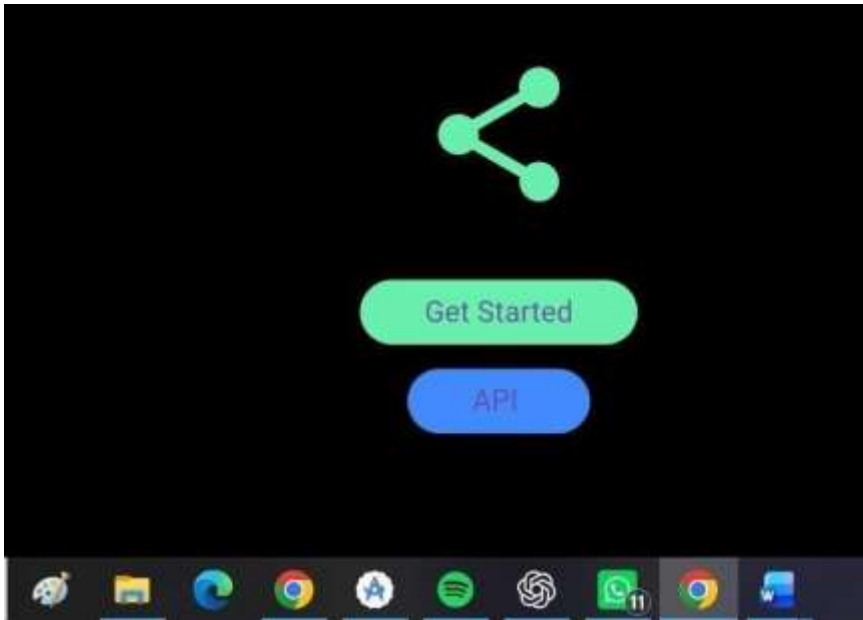


Figure 2

This shows when the internet access is denied, the server cannot be reached.

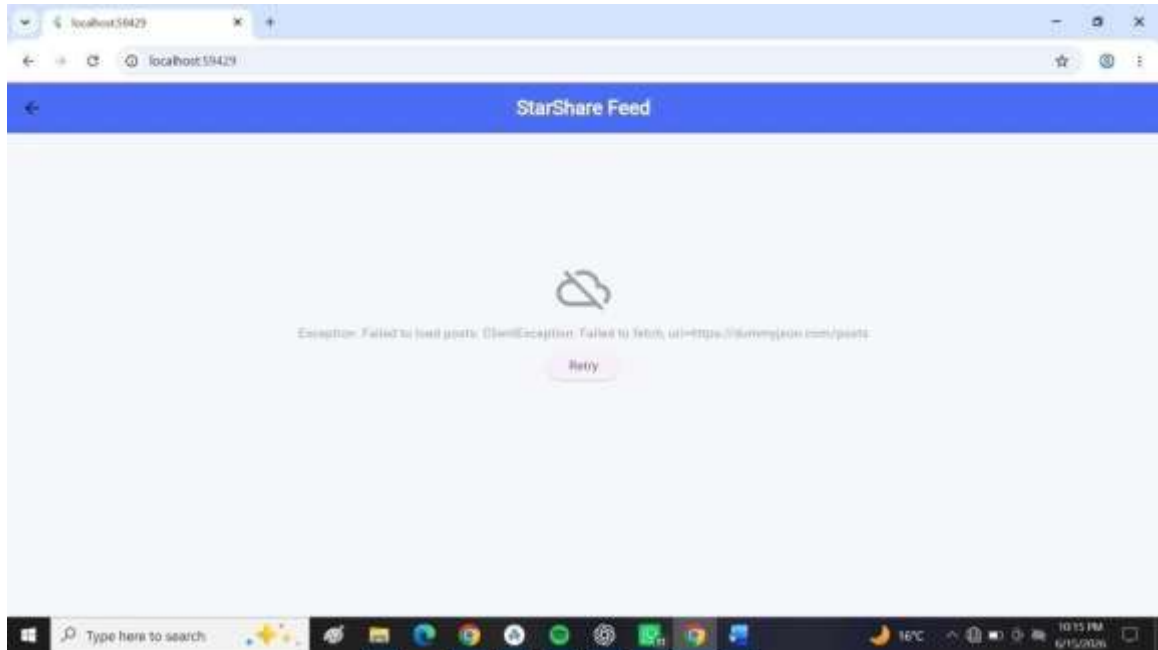
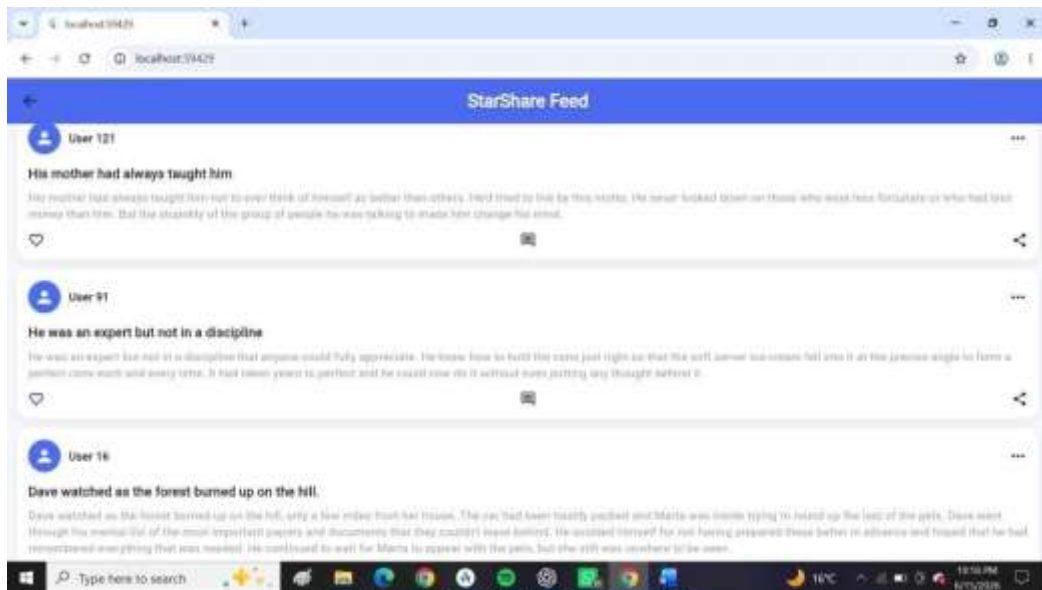


Figure 3

1. Handled API responses and the internet error and reached the information.



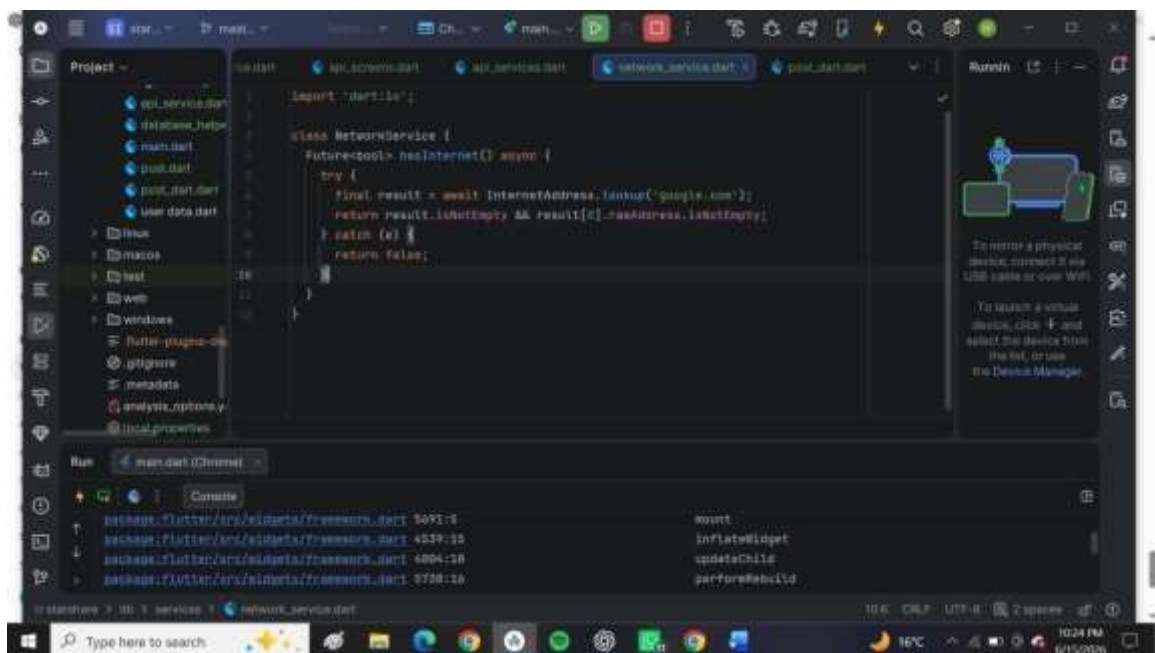
Tools

1. Flutter SDK
2. Dart
3. HTTP package
4. Backend API (Node.js / PHP / Firebase if used)
5. Android Studio

Personal Reflection

I learned how to connect a mobile application to external APIs and manage real-time data exchange. This improved my understanding of networking, asynchronous programming, and backend integration.

The network connection string code



Why does my system (Starshare) require networking?

StarShare requires networking to upload, sync, and retrieve files and user data between devices and the server.

Week 7: Application Architecture

Same as week 1

Application: StarShare File Sharing App

TASK 1: STORAGE METHOD COMPARISON (STARSHARE)

StarShare uses different storage methods depending on functionality.

Storage Method	Usage in StarShare	Advantages	Disadvantages
Shared Preferences	Stores user login status	Fast and simple	Not secure for sensitive data
SQLite Database	Stores file transfer history	Works offline and structured	Requires SQL knowledge
Firebase	Handles user authentication	Secure and real-time	Requires internet connection
Internal Storage	Stores received files	Private and offline access	Limited storage space

TASK 2: DATABASE DESIGN (STARSHARE SYSTEM)

System Overview

StarShare is a mobile application used for file sharing between devices using WiFi connection.

USERS TABLE (FIREBASE AUTH)

Field	Type	Description
user_id	STRING	Unique Firebase user ID
Field	Type	Description
email	STRING	User email address
password	STRING	Managed securely by Firebase Authentication

TRANSFERS TABLE (LOCAL SQLITE DATABASE)

Field	Type	Description
transfer_id	INTEGER	Primary key
file_name	TEXT	Name of file transferred
file_size	INTEGER	Size of file in bytes
direction	TEXT	SEND or RECEIVE
ip_address	TEXT	Device IP address
timestamp	TEXT	Date and time of transfer

SQL IMPLEMENTATION

```
CREATE TABLE Transfers ( transfer_id INTEGER PRIMARY KEY
AUTOINCREMENT,
file_name TEXT, file_size
INTEGER, direction TEXT,
ip_address TEXT, timestamp
TEXT
);
```

RELATIONSHIP DESIGN

Each user can perform multiple file transfers.

Relationship:

User (1) ----- (Many) Transfers

NORMALIZATION

First Normal Form (1NF):

- All fields contain atomic values

Second Normal Form (2NF):

- All attributes depend on the primary key

Third Normal Form (3NF):

- No transitive dependencies exist

PASSWORD SECURITY (FIREBASE)

StarShare uses Firebase Authentication.

- Passwords are NOT stored manually
- Firebase automatically encrypts and hashes passwords

Example:

Password: 123456

Stored: Encrypted hash value (Firebase managed)

TASK 3: MINI PROJECT (STARSHARE SYSTEM DESIGN)

Storage Methods Used

StarShare uses a hybrid storage system:

- Firebase Authentication
- SQLite Database
- Internal Storage

REASONS FOR SELECTION

Firebase:

- User authentication
- Secure login system
- Cloud-based storage support **SQLite:**
- Stores file transfer history
- Works offline

Internal Storage:

- Stores received files locally on device

DATABASE STRUCTURE

Users Table (Firebase)

- user_id
- email

Transfers Table (SQLite)

- transfer_id
- file_name
- file_size
- direction
- ip_address
- timestamp

EXPECTED OUTPUT

The StarShare application should:

- Allow user registration and login • Discover nearby devices
- Send files using WiFi connection
- Receive and store files locally
- Save transfer history
- Display sent and received file logs

CONCLUSION

StarShare demonstrates real-world mobile application networking using:

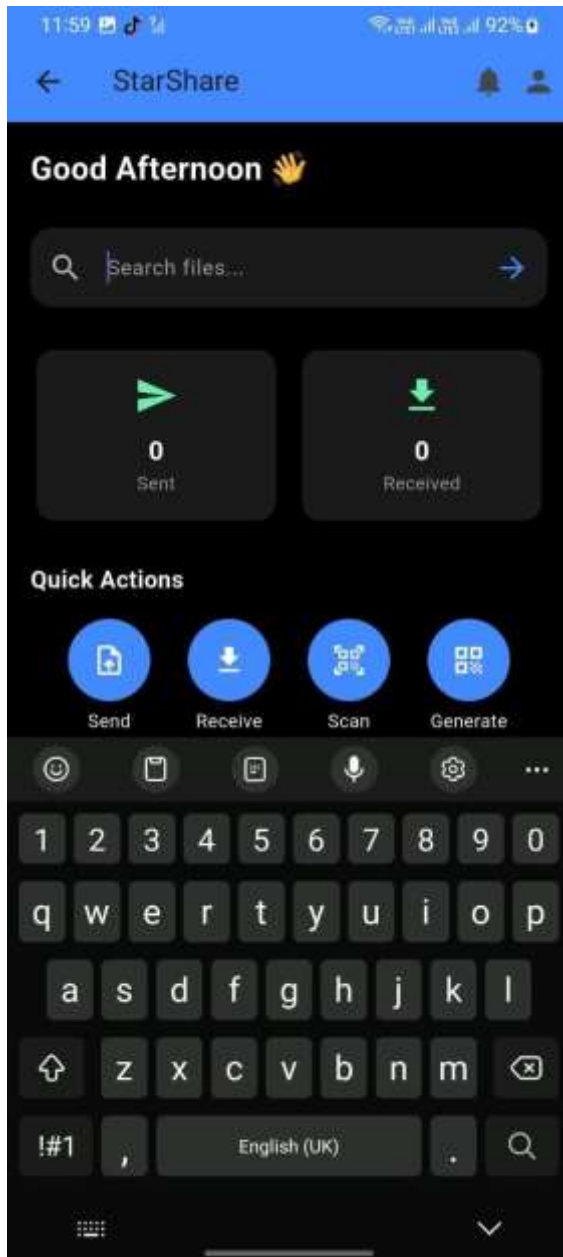
- Client-server architecture (TCP sockets)
- Device discovery (UDP broadcast)
- Cloud authentication (Firebase)
- Local data storage (SQLite)

Week 8:

Task 1: Keyboard Input Implemented:

- Search TextField
- Input validation using InputValidator
- SnackBar for empty search
- Keyboard hides after search

Figure 1



The screenshot above proves the keyboard inputs where a user can enter the name of the file and search it.

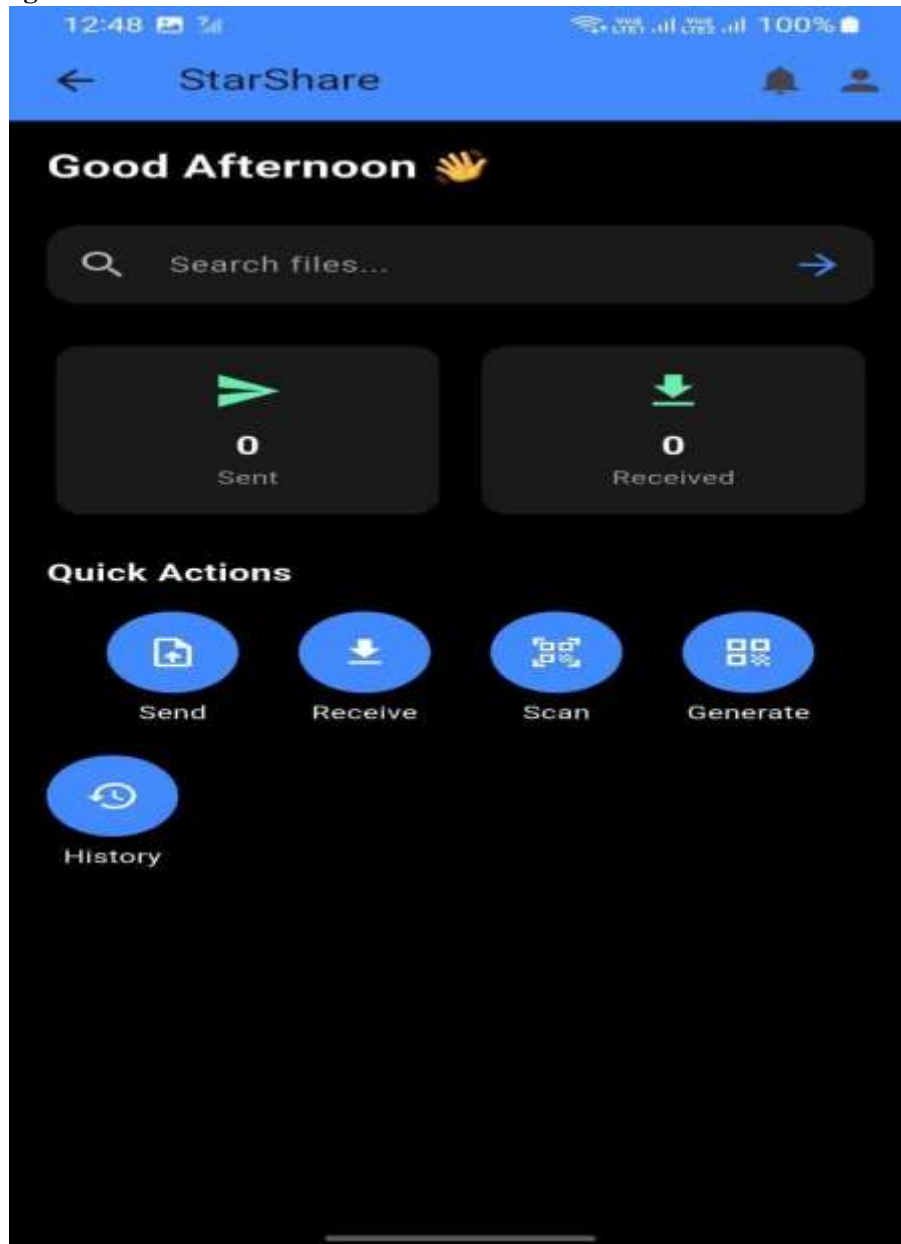
Task 2: Tap Events

Implemented separate methods for handling user actions:

- Send
- Receive

- Scan/Connect
- Generate QR
- History

Figure 2



Task 3: Long Press Write Task 3: Long Press Gesture

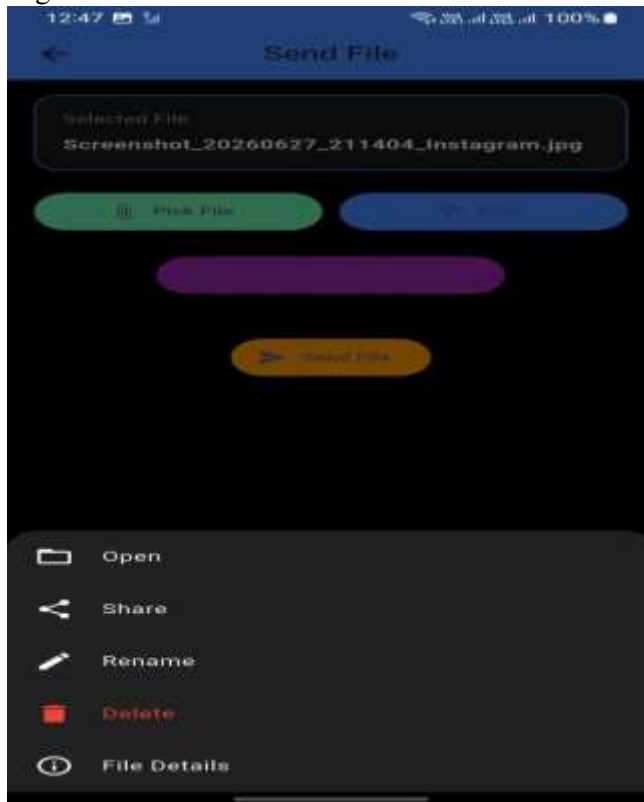
Implemented long press on the Selected File card.

Options displayed:

- Open
- Share
- Rename
- Delete
- File Details

Implemented using the GestureHandler class.

Figure 3.



NB: When selected file is long pressed, the above options are provided at the bottom of the screen.

Task 4: Swipe Down Gesture

Implemented swipe down to refresh nearby devices.

Calls:

- discoverDevices()
- Displays SnackBar "Refreshing nearby devices..." **Figure 4.**



When user swipes down, the list of present devices are refreshed.

Week 9: Device Features and Sensors

Week 9: Device Features Integration (Camera, GPS and Sensors)

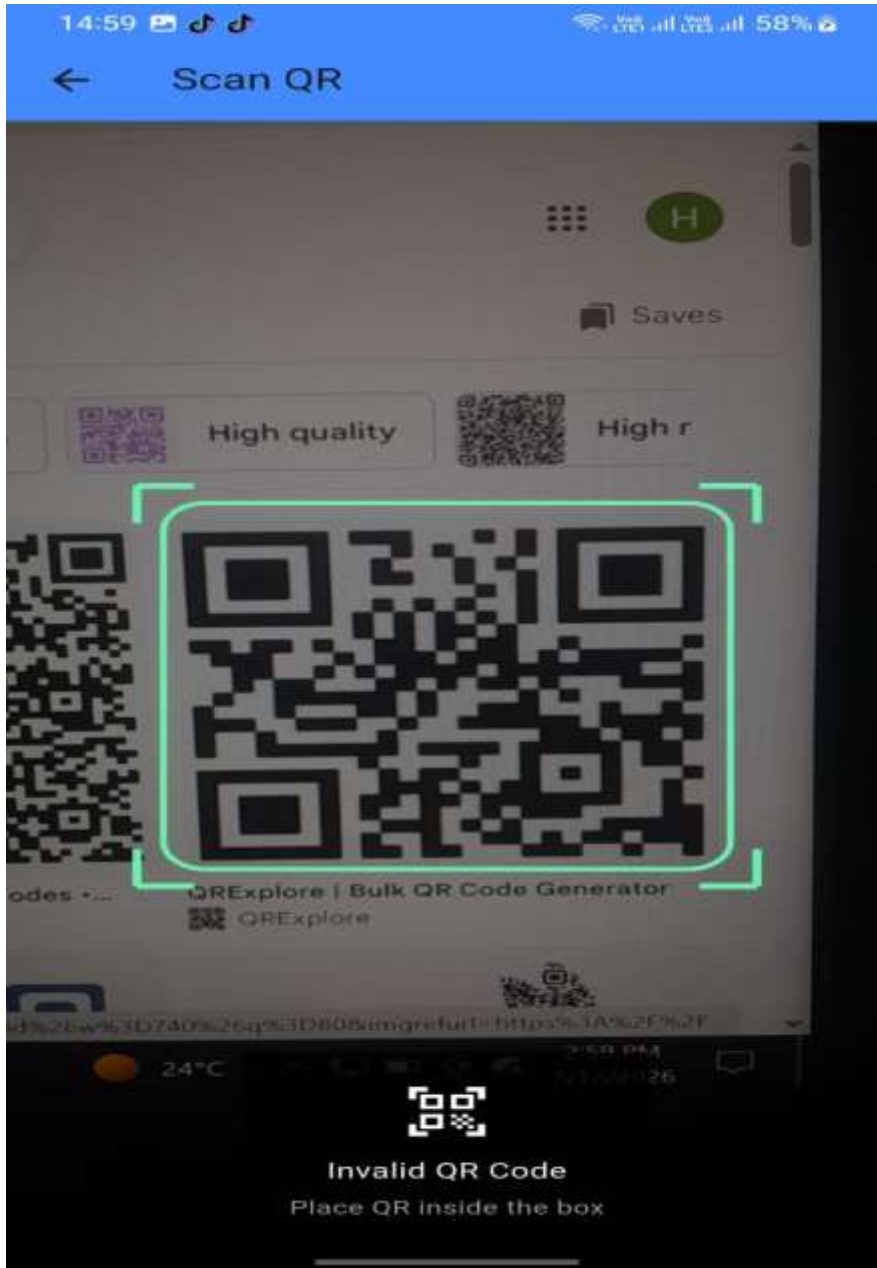
Objective

To integrate device hardware features into the StarShare application by implementing camera access for QR code scanning, GPS location services, and runtime permissions.

Activity 1: Camera Integration

Description

Implemented camera functionality using the **Mobile Scanner** package. The camera scans StarShare QR codes, extracts the device IP address and port number, and prepares the application for secure file transfer.



Explanation

Figure 1: The StarShare QR Scanner uses the device camera to scan QR codes and automatically retrieve connection details.

Activity 2: QR Code Generation

Description

Implemented QR code generation to allow other StarShare devices to connect easily. The generated QR code contains the device IP address and communication port.



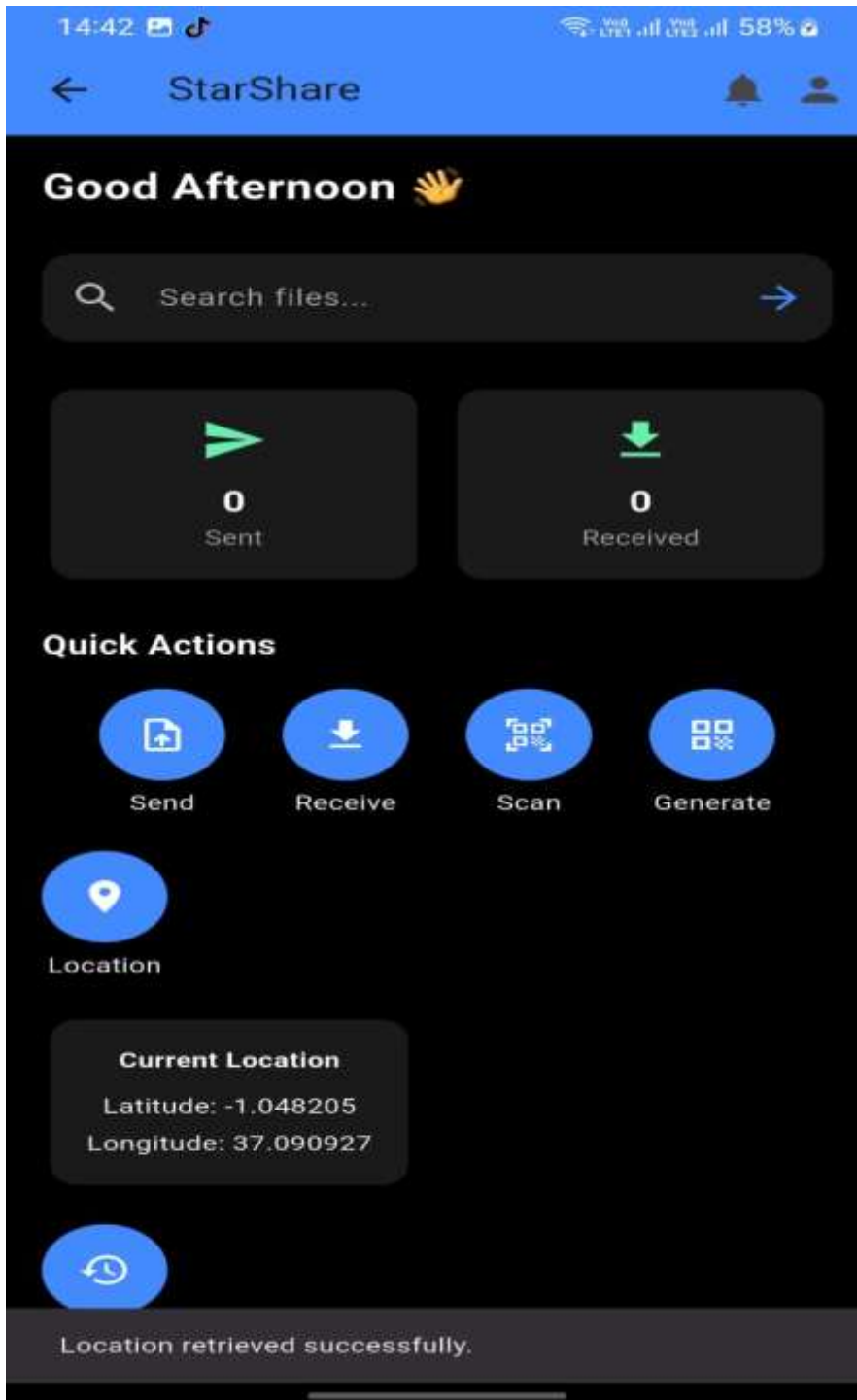
Explanation

Figure 2: The generated QR code contains the device IP address and port used for establishing a connection.

Activity 3: GPS Integration

Description

Integrated GPS functionality using the Geolocator package. The application retrieves the user's current latitude and longitude after requesting location permission.

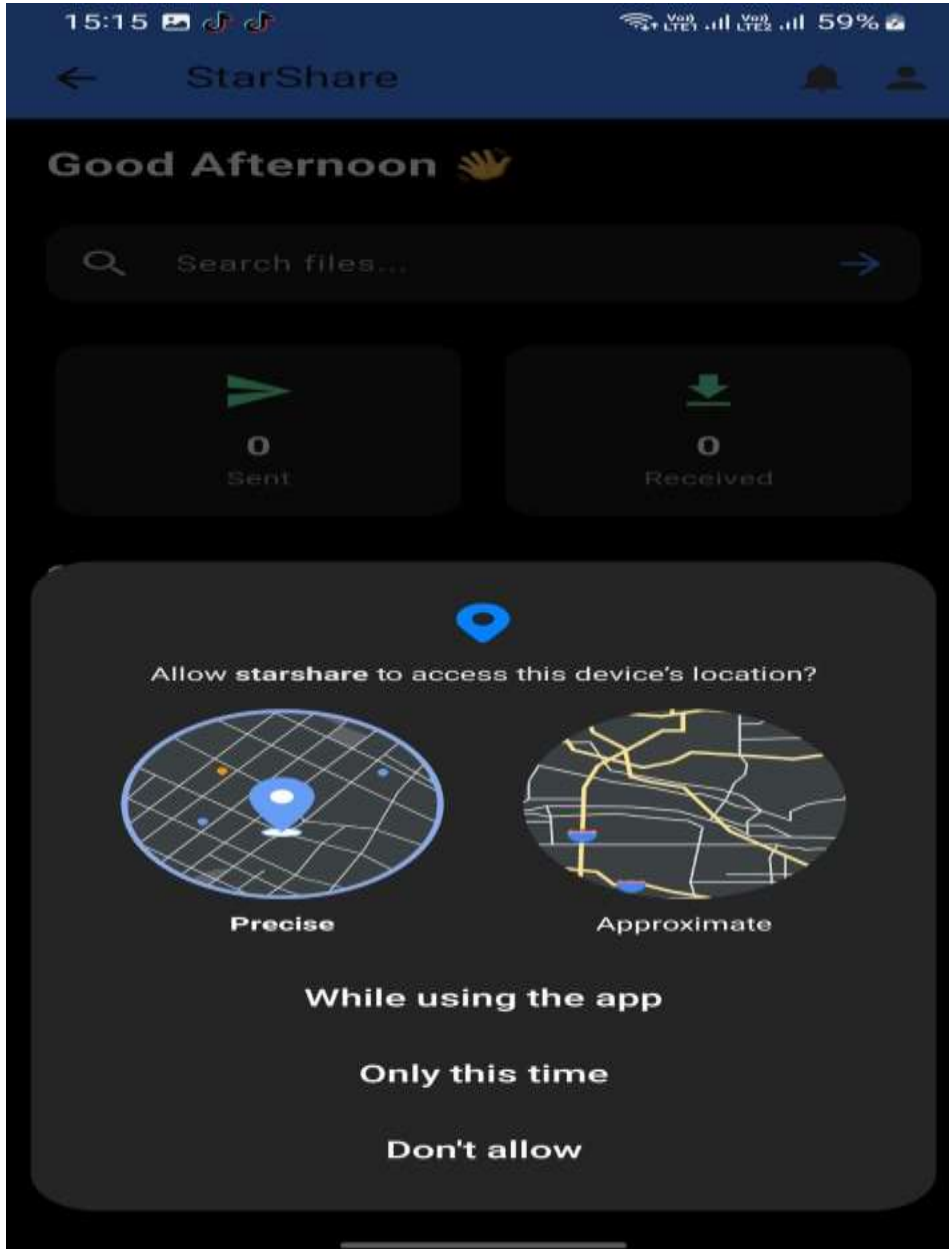


Explanation

Figure 3: The application successfully retrieves and displays the current GPS coordinates of the device.

Activity 4: Runtime Permissions

Implemented Android runtime permission handling to request user authorization before accessing location services.

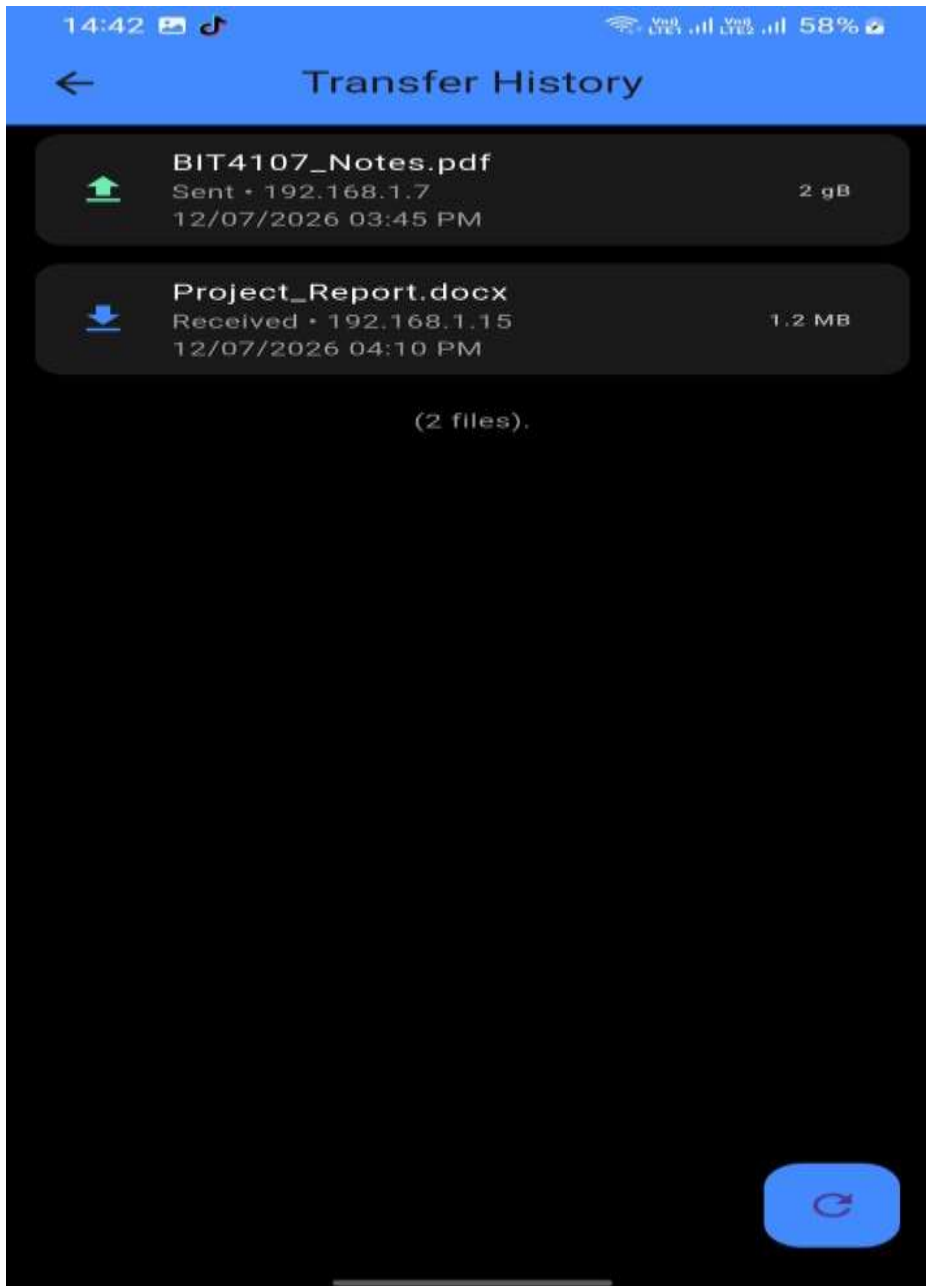


Explanation

Figure.4: The application requests permission before accessing the device location to protect user privacy.

Activity 5: Transfer History

Updated the History screen to display transferred files. A sample transfer record is displayed during development to demonstrate the user interface before actual file transfers are performed.



Explanation

Figure 5: The History screen displays details of transferred files including file name, device IP address, file size, and timestamp.

Challenges Encountered

- Configuring Android location permissions.
- Requesting runtime permissions correctly.

- Integrating GPS without affecting the existing application.
- Testing the QR scanner and GPS functionality.

Solutions Implemented

- Added the required permissions in the AndroidManifest.xml file.
- Created a Location Service using the Geolocator package.
- Added a Location button to display the current coordinates.
- Successfully integrated QR code scanning using the device camera.

Learning Outcomes

By the end of this practical session, I was able to:

- Integrate the device camera into a Flutter application.
- Generate and scan QR codes.
- Retrieve the current GPS location.
- Request and manage Android runtime permissions.
- Display location information within the application.
- Enhance the StarShare application using Android device features.

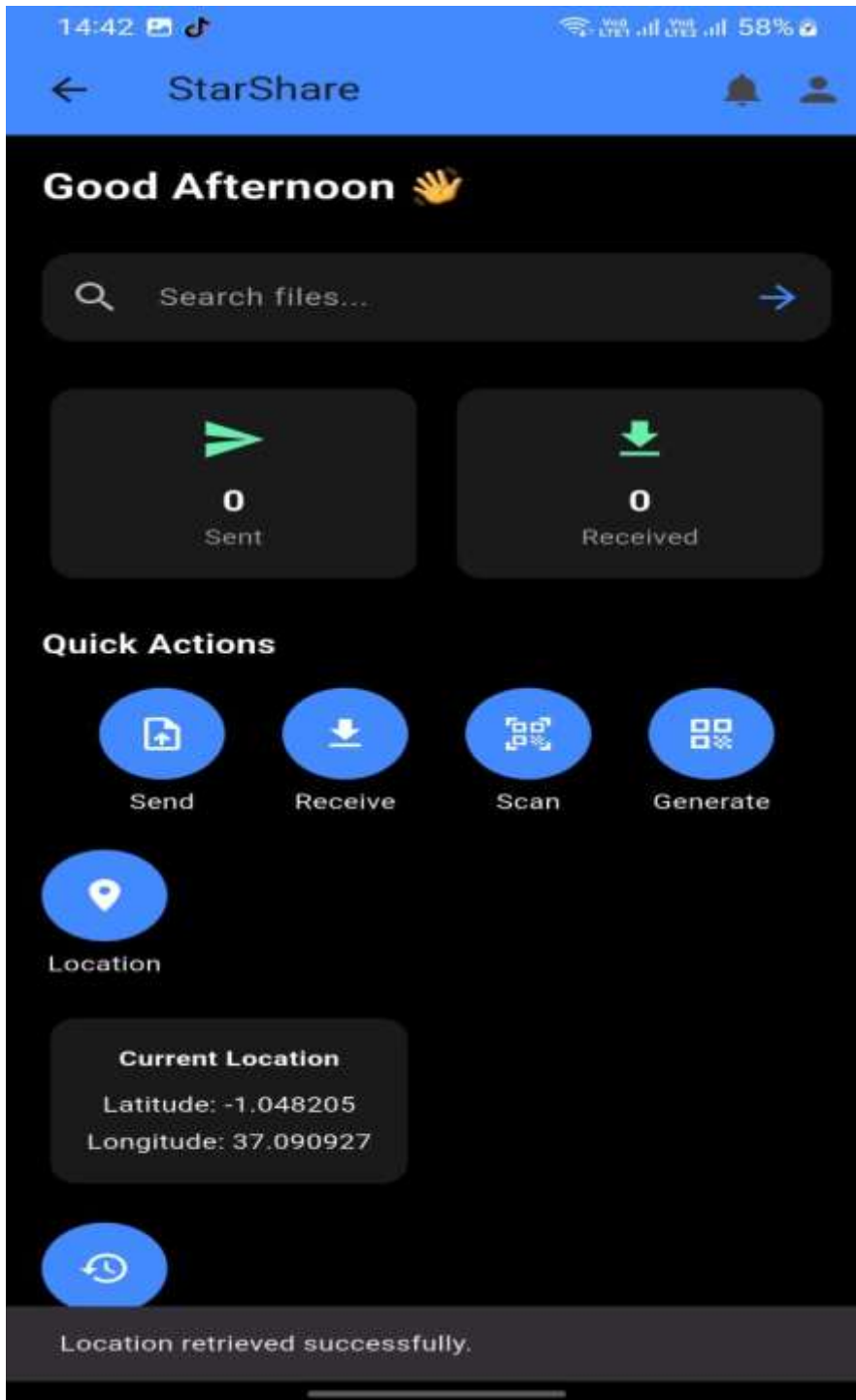
Week 10: Notifications and Background Services

WEEK 10 – Comprehensive Application Development, Integration and Testing

Activity 1: Application Integration

Description

Integrated all previously developed modules into one complete application. The Login, Home, QR Scanner, QR Generator, Send File, Receive File, History, and GPS features were connected to ensure smooth navigation and functionality.



Home Screen showing all quick action buttons.

Figure 1: Integrated StarShare Home Screen with all application modules connected.

Activity 2: Functional Testing

Description

Tested all major features of the application to verify that they function correctly after integration.

Feature	Status
Login	Passed
Navigation	Passed
QR Scanner	Passed
QR Generator	Passed
GPS Location	Passed
Search Function	Passed
Transfer History	Passed



Figure 2: Functional testing confirming successful operation of integrated features such as the qr generator screen

Activity 3: Bug Fixing and Improvements

Description

Resolved issues encountered during integration, including GPS permission handling, QR scanner navigation, transfer history updates, and user interface improvements. Search functionality and location display were also enhanced.

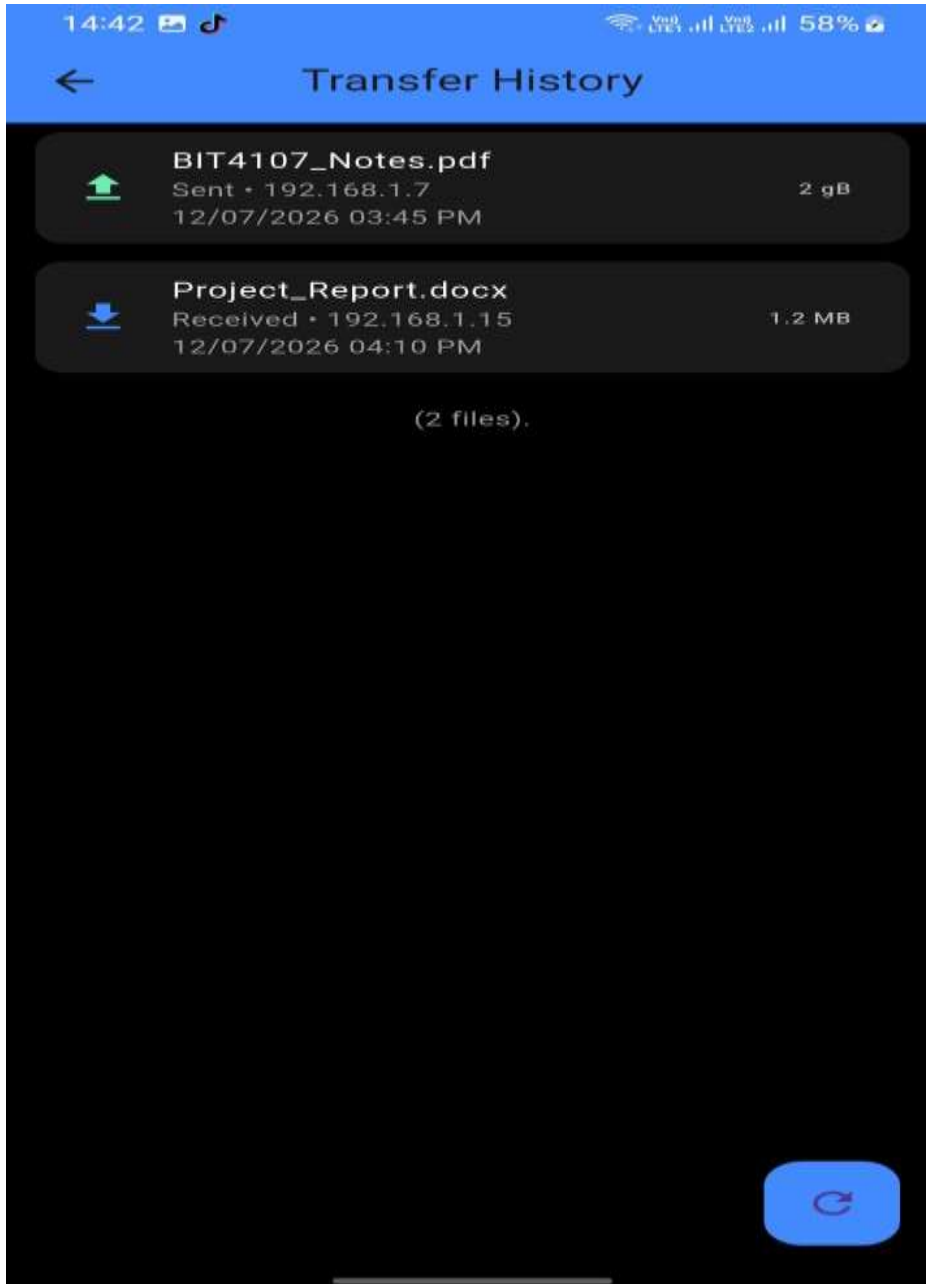


Figure 3: Updated StarShare history screen interface after bug fixes and feature improvements.

Week 10 Learning Outcome

Successfully integrated all application modules into one complete mobile application, performed functional testing, fixed identified bugs, and improved overall application usability.

Week 11: Application Deployment and Publishing

WEEK 11 – Application Deployment and Publishing

Activity 1: Application Preparation

Description

Prepared the StarShare application for deployment by reviewing the user interface, verifying permissions, testing navigation, and ensuring all features operated correctly.

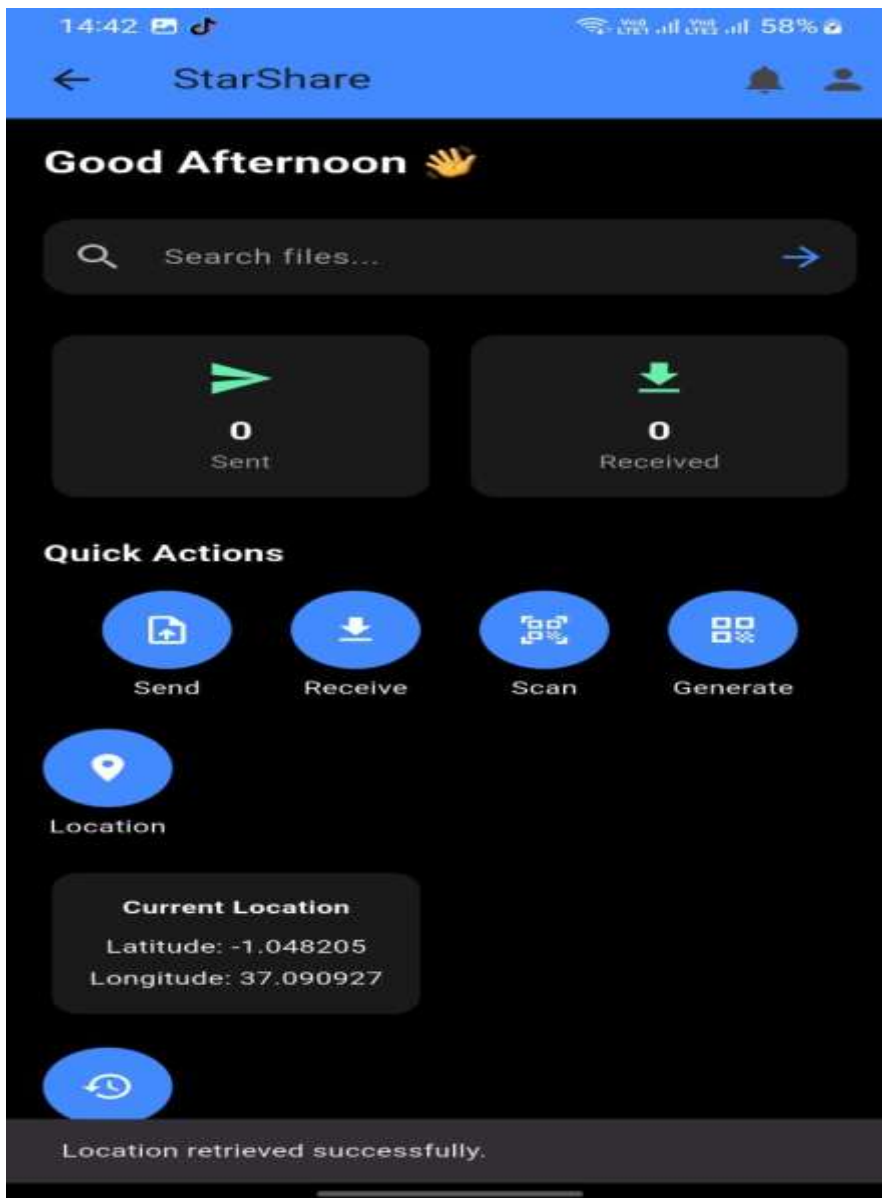


Figure 11.1: Final version of the StarShare application home screen ready for deployment.

Activity 2: Release APK Generation

Description

Generated the release APK using Flutter to prepare the application for installation and distribution on Android devices.

Command Used

```
flutter build apk --release
```

Terminal showing successful APK generation.

Figure 11.2: Successful generation of the StarShare release APK.

Activity 3: Final Application Documentation

Description

Prepared the application for presentation by organizing screenshots, documenting implemented features, and confirming readiness for deployment.

Week 11 Learning Outcome

Successfully prepared the StarShare application for deployment by generating a release APK, performing final testing, and organizing project documentation.

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic Major Challenges Faced How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: Haron Mutai

Student Signature:

Date: 9th June 2026

Lecturer Name: Michael Nyoro

Lecturer Remarks: _____

Date: _____